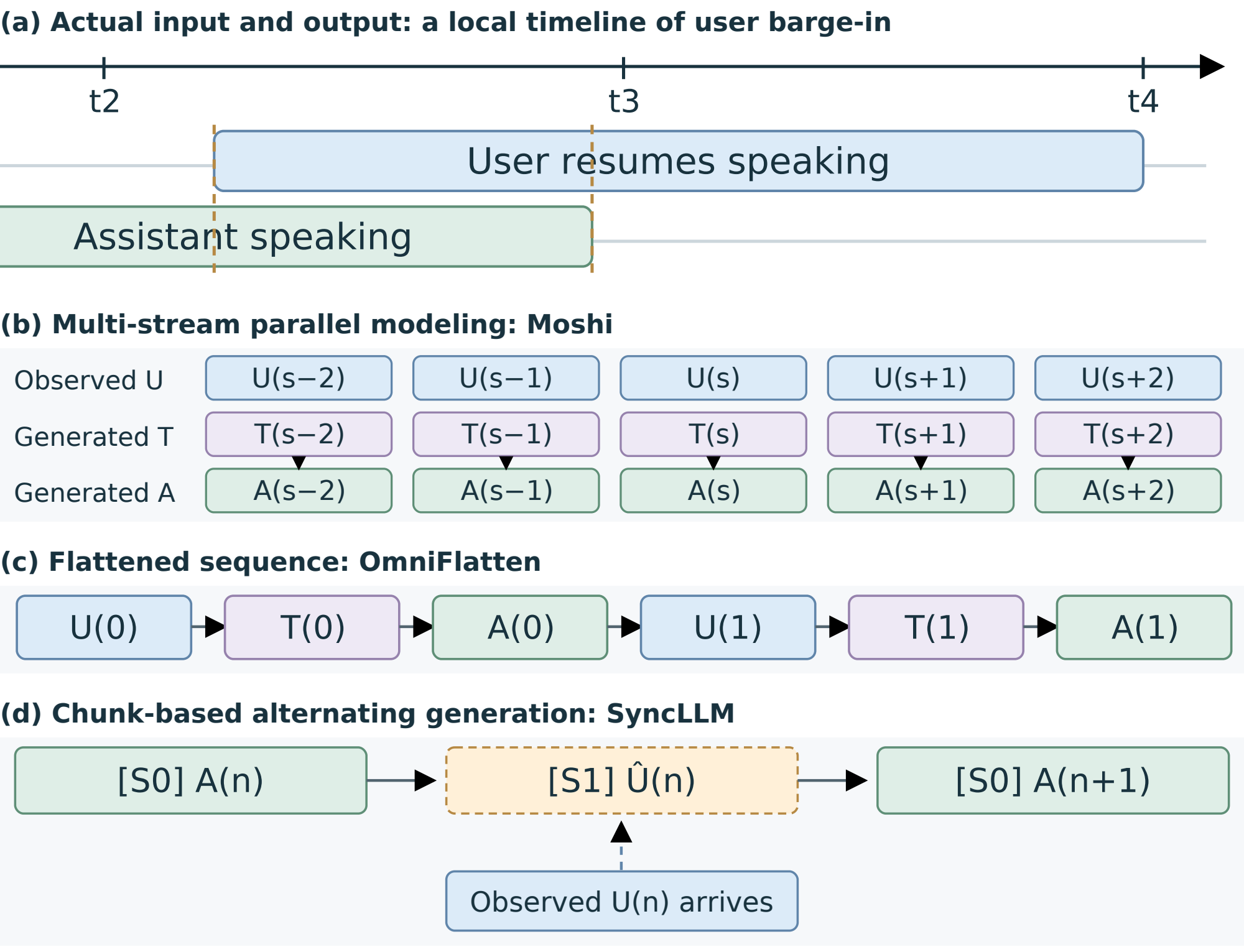
(a) Actual input and output: a local timeline of user barge-in
t2
t3
t4
User resumes speaking
Assistant speaking
(b) Multi-stream parallel modeling: Moshi
Observed U
Generated T
Generated A
U(s−2)
U(s−1)
U(s)
U(s+1)
U(s+2)
T(s−2)
T(s−1)
T(s)
T(s+1)
T(s+2)
A(s−2)
A(s−1)
A(s)
A(s+1)
A(s+2)
(c) Flattened sequence: OmniFlatten
U(0)
T(0)
A(0)
U(1)
T(1)
A(1)
(d) Chunk-based alternating generation: SyncLLM
[S0] A(n)
[S1] Û(n)
[S0] A(n+1)
Observed U(n) arrives